

Curriculum Vitae
Emily C. Church

University of Illinois at Chicago
Graduate Program in Neuroscience
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EDUCATION & TRAINING

- University of Illinois at Chicago* GPA: 3.83
Enrolled in Neuroscience Ph.D. program 2009
Passed preliminary exam without modifications 2011
Passed defense 2020
- École des Neurosciences Paris Île-de-France*
“Optical Imaging and Electrophysiology Methods in Neuroscience”
2013 Spring Course
- Boston University*
“An Introduction to Evidence-Based Undergraduate Stem Teaching”
2015 Massive Open Online Course
- University of Wisconsin-Madison* GPA: 3.58
Graduated with dual degrees, May 2008
- Bachelor of Science: Natural Sciences—Nutritional Sciences
 - Bachelor of Science: Kinesiology—Exercise and Movement Science
- Century Community and Technical College* GPA: 4.00
White Bear Lake, MN
Post Secondary Enrollment Option for gifted high school students
Earned high school and college credit simultaneously, Fall 2002—Spring 2003

PUBLICATIONS

- Church E**, Hamid E, Zurawski ZP, Alford S. Synaptic integration by co-localized G protein-coupled receptors in presynaptic terminals. *BioRxiv* (2020).
- Hamid E, **Church E**, Alford S. Quantitation and simulation of single action-potential-evoked Ca^{2+} signals in CA1 pyramidal neuron presynaptic terminals. *eNeuro* (2019).
- Zurawski Z, Gray ADT, Brady LJ, Page B, **Church E**, Harris NA, Dohn MR, Yim YY, Hyde K, Mortlock DP, Jones CK, Winder DG, Alford S, Hamm HE. Disabling the $\text{G}\beta\gamma$ -SNARE interaction disrupts GPCR-mediated presynaptic inhibition, leading to physiological and behavioral phenotypes. *Sci Signal* (2019).
- Hamid E, **Church EC**, Wells C, Zurawski Z, Hamm HE and Alford S. Modulation of neurotransmission by GPCRs is dependent upon the micro-architecture of the primed vesicle complex. *J. Neurosci* (2014). PubMedID#24381287
- Goslawski M, Piano MR, Bian JT, **Church E**, Szczurek M, Phillips SA. Binge drinking impairs vascular function in young adults. *J Am Coll Cardiol* (2013), doi: 10.1016/j.jacc.2013.03.049
- Varady KA, Bhutani S, **Church EC**, Phillips SA. 2010. Adipokine responses to acute resistance exercise in trained and untrained men. *Med Sci Sports Exerc* 42(3):456-62. PubMedID#19952805
- Phillips, SA; Bian, JT; **Church, EC**; Das, EK; Vidovich, M; Gutterman, DD. 2009. Hydrogen Peroxide Prevents Impaired Endothelium-dependent Dilation Following Acute Exertion in Chronic Exercising but Not in Sedentary Subjects. *Circulation* 120(18):S1013. Accession Number: WOS:000271831503460

Varady KA, Bhutani, S, **Church EC**, Klempel MC. 2009. Short-term modified alternate-day fasting: A novel dietary strategy for weight loss and cardioprotection in obese adults. *Am J Clin Nutr* 90(5):1138-43. PubMedID#19793855

ABSTRACTS

Church EC, Brown J, Alford S. Proper Statistical Analyses of Electrophysiological Data. Poster session presented at 45th Annual Society for Neuroscience Conference: 2015 Oct 17-21; Chicago, IL.

Church EC, Hamid E, Alford S. Monte Carlo simulation of transmitter release and receptor activation at CA1 glutamatergic synapses supports modified fusion dynamics as a mechanism of 5-HT_{1B}R inhibition. Dynamic poster session presented at 44th Annual Society for Neuroscience Conference: 2014 Nov 14-19; Washington, DC.

Church EC, Hamid E, Alford S. Presynaptic inhibition of neurotransmitter release in mammalian hippocampal synapses. Poster session presented at 43rd Annual Society for Neuroscience Conference; 2013 Nov 9-13; San Diego, CA.

Church EC, Rivero-Covelo I, Braun DJ, Fettiplace MR, Cone JJ, Baker PM, Alpert MH. Lessons learned in the development of a primary literature-based undergraduate neuroscience course. Theme H poster session submitted at 43rd Annual Society for Neuroscience Conference; 2013 Nov 9-13; San Diego, CA.

Church EC, Gong LW, Alford S. 5-HT_{1B} and α_{2A} -AR functionally expressed in hippocampal culture. Poster session presented at 42nd Annual Society for Neuroscience Conference; 2012 Oct 13-17; New Orleans, LA and Chicago Chapter 2013 Annual Meeting; Mar 7; Chicago, IL.

Varady KA, Bhutani S, **Church EC**, & Klempel MC 2010. Alternate day fasting: A novel dietary strategy for weight loss and cardio-protection in obese adults. *The FASEB Journal* (Vol. 24). 9650 ROCKVILLE PIKE, BETHESDA, MD 20814-3998 USA: FEDERATION AMER SOC EXP BIOL.

Varady KA, Bhutani S, **Church EC**, Klempel MC. Acute effects of weight lifting on plasma adiponectin in trained versus untrained individuals. 2009. *The FASEB Journal* (Vol. 23). LB451

Koenig, R., Dickman, J., **Church, E.**, Umfress, R., & Ji, L. L. 2009. Effect Of Avenanthramide Supplementation On Responses To Eccentric Exercise In Young Women: 2088: Board# 229 *Medicine & Science in Sports & Exercise*, 41(5), 228-229.

Feng H, Awoyinka L, Kang CH, Dickman JR, Koenig R, **Church EC**, Gu L, Ji LL. 2009. Nuclear Factor Kappa B And Mitochondrial Biogenesis: Effect Of Endurance Training: 720: *Medicine & Science in Sports & Exercise*, 41(5), 61.

FELLOWSHIPS & AWARDS

W.C. and May Preble Deiss Fund for Biomedical Research Award: 2014

1st Place Neuroscience Day Poster Session- Neuroscience Day Travel Award: 2014

Teaching Excellence Award: 2014

Chancellor's Graduate Research Fellowship: 2014, 2015

Training Grant in the Neuroscience of Mental Health (T32MH067631): 2013-2014 Academic Year

Abraham Lincoln Fellowship: 2012-2013 Academic Year

Graduate Student Council Travel Award: 2012, 2013

Graduate College Presenter's Award: 2012

Biological Sciences Travel Award: 2012, 2013

Neuroscience Graduate College Fellowship: 2009-2010 Academic Year

Peter Young Student Assistant Grant: 2007-2008 Academic Year

William F. Vilas Grant: 2006-2007

Dean's List: Fall 2003, Fall 2004, Summer 2007

Dean's Honor List: Spring 2005, Spring 2006, Fall 2006, Spring 2007, Fall 2007, Spring 2008

RESEARCH/LABORATORY EXPERIENCE

Graduate Student, UIC Neuroscience: Alford Lab, University of Illinois at Chicago (9/2010-present)
Study G protein-receptor-mediated modulation of synaptic plasticity using electrophysiology and imaging in rodent hippocampus. Develop neural culture and transfection protocol for electrophysiology, binding and fusion assays.

Visiting Research Specialist: Phillips Lab, University of IL at Chicago (9/2008-9/2009)
Oversee clinical study of exercise effects on vascular health. Recruit, screen, and schedule research subjects. Coordinate with CRC nurses, MDs, and students. Obtain informed consent from research subjects. Collect brachial artery diameter and flow data with Doppler ultrasound. Obtain blood pressure, heart rate, and O₂ saturation data during resistance exercise. Assist MD/APN with gluteal fat biopsy. Analyze data and blood. Work with Human Subjects IRB to maintain current and accurate approvals. Completed HIPAA training and certification.

Clinical Coordinator: Varady Lab, University of IL at Chicago (2/2009-6/2009)
Recruit and coordinate weight loss clinical research feeding study on obese men and women. Provide dietary counseling and take weekly anthropometric measurements on subject volunteers.

Chemistry Lab Tech: Ji Lab, UW-Madison (6/2008—8/2008)
Use western blots to determine role of specific proteins in antioxidant cascade response from exercise-trained rats.

Lab Assistant/Associate Research Specialist: Valdivia Lab, UW-Madison (9/2006-8/2008)
Wean, tail clip, ear punch, inventory, and euthanize transgenic mice. Mouse tail digestion, DNA isolation, PCR, and gel electrophoresis. Make solutions, place orders, general lab maintenance. Train undergraduate assistant. Perform surgery on mice, isolate adult mouse cardiomyocytes using Langendorff perfusion. Measure calcium transients using confocal microscopy.

Independent Project: under Dr. Héctor Valdivia, Madison, WI (Spring Semester 2008)
Analyze cardiomyocytes from mice with intracellular calcium handling mutations using western blots.

Kinesiology Practicum: under Dr. Li Li Ji, UW-Madison (Spring Semester 2008)
Assist with a project on the effects of oat antioxidant supplementation on eccentric exercise stress in young women. Analyze subjects' blood for antioxidants and reactive oxygen species (ROS).

Lab Assistant, Plant Disease Diagnostics Clinic, Madison, WI (5/2006-8/2006)
Bacterial and fungal test various plant species, prepare solutions and media, update national database

TEACHING

Invited Lecture at North Eastern Illinois University

Research talk with an emphasis on statistics to upperclassman and master's students in Cindy Voisine's neurobiology class at NEIU (2016)

BIOS 386: Advanced Topics in Modern Neuroscience

Created a novel undergraduate course in collaboration with two fellow GPN students for the purpose of teaching critical thinking skills and the scientific process to undergraduates while providing GPN students with meaningful teaching experience. Served as course coordinator and co-instructor. (2012-2014)

NEUS 501: Fundamentals of Neuroscience

Led a post-exam review session on G protein-coupled receptors for first-year graduate students. (2012)

KINES 318: Biomechanics of Human Movement

Spring 2008 tutor.

ATTENDANCE AT CONFERENCES & COLLOQUIA

ComSciCon Chicago 2015

Society for Neuroscience Annual Conference: 2014, 2013, 2012, 2010, 2009

Chicago Chapter Society for Neuroscience Annual Conference: 2013, 2012, 2010, 2009

Brain Research Foundation-Neuroscience Day: 2013 presentation

CBC Scholars 'Loop Connections' Colloquium: 2012

TRAINEES

Neemah BassiriRad; Technician

Gajanan Kulkarni, Undergraduate Assistant

Saba Rezaeian, Undergraduate Assistant

Melissa Goslawski; M.S., M.D.

Jessica Pipal; P.T.

Tselmeg Ulzibayar; Undergraduate Assistant

Holli Loomans; Laboratory Technician

VOLUNTEERING

Work + Shelter Advisory Board member (2016-present)

Chicago North Co-Leader for Citizens' Climate Lobby (2013-present)

Registration & Workshop Volunteer for Expanding Your Horizons (March 2016)

Volunteer for CDOT Bicycle Program (March 2012)

Brain Bee tour leader—Presented laboratory tours to high school students (February 2012)

Volunteer for Sierra Club Air & Energy Committee (2009-2012)

Judge—Undergraduate Neuroscience Poster Session (2011)

Group leader for Expanding Your Horizons (EYH)—A program for middle school girls interested in science and math (November 2006 and 2007)

Volunteer for Planned Parenthood and Fair Wisconsin (2006)

Volunteer for Smoke-Free Madison (2004)

Volunteer as research subject for various scientific studies (2004-2012)

SKILLS

Whole-Cell Patch Clamp, Acute and Primary Neural Culture, Organotypic Culture, Live Cell Imaging, Mouse Husbandry, Mouse and Rat Dissection and Surgery, Confocal Microscopy, TIRF Microscopy, Two-photon Microscopy, Gene Transformation and Transfection, PCR, Gel Electrophoresis, ICC, IHC, Western Blotting, Spectrophotometry, Doppler Ultrasound, Brachial Analyzer, Axograph, pCLAMP, Igor, ImageJ, Adobe Illustrator, R